TERM PROJECT PROPOSAL · 1.125 AGENTIC ENGINEERING

StarCrew



An AI agent that plans — and re-plans — **group stargazing trips**

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1 · PROBLEM AND AUDIENCE

Group stargazing is chaos to plan

A good night needs all of these to line up:

- Clear sky
- No bright Moon
- Dark site
- Right hours
- Everyone free
- Rides sorted

Today that means checking weather models, Moon tables, light-pollution maps and satellite trackers — then negotiating it all in a group chat.

> *"I've been in stargazing groups before. Trips kept falling apart: nobody could line up the forecast, the Moon, the site and everyone's schedule."*

Group chat · illustrative

Is Saturday clear? Apps disagree…

Full Moon this week next week?

Who has a car? I can only do Friday.

Clouds rolling in… cancel?

Audience: astronomy clubs, campus groups, friends chasing a meteor shower

2 · PROPOSED SOLUTION

One request in, a whole trip plan out

“Our group of 5 wants to see the Geminids in December — max 2 hours’ drive from Boston.”

01 Gather

Each member’s start point, free nights and car seats

02 Plan

Best night, dark site and time window, with a scorecard and confidence level

03 Share

Sky chart, what to look for and when, and a ready-to-send group message with carpools

04 Watch

Re-checks the forecast before departure and proposes a backup site or night

EXAMPLE OUTPUT · MOCK-UP

Night	Site	Window	Clouds	Confidence
[Sat, Dec __]	**[Dark site X]**	**[21:30–02:00]**	**3 models agree**	**High**

3 · APPROACH

The LLM reasons; deterministic tools supply the facts

Group request + members' constraints

↓

Planner agent

Claude API · tool use · plan → call → check

↓

Plan + scorecard + sky chart + group message

↻ Day-of re-check → re-plan if conditions change

→

Weather

Open-Meteo (multi-model clouds), 7Timer! seeing

Astronomy

Skyfield + JPL ephemeris: Moon, darkness, planets

Satellites & meteors

CelesTrak orbits (ISS passes), meteor-shower calendar

Dark sites

Curated Boston-area sites + light-pollution data

Routing

Drive times from each member's start point

Group solver

Checks availability, seats and max-drive limits

Every number traces back to a tool — never to the LLM

Stack: Python · FastAPI · Claude API · Skyfield · web UI (map + charts)

4 · SCOPE AND RISKS

Boston first, solo first, groups next

MVP · BY MIDPOINT

Solo planner: 10–20 Boston-area sites, next 7 nights, cloud confidence

FULL · BY FINAL

Group coordination, day-of re-check and re-plan, group message

STRETCH · NOT PROMISED

Sites in China, star labels on photos, chat-bot integration

Risk	Level	Mitigation
Cloud forecasts are uncertain	Medium	Compare models → confidence level; re-check on the day; backtest on history
Light-pollution data is hard to query	Medium	Start from a curated site list with darkness ratings
Group logic gets complex	Medium	Groups of 3–6; hard constraints first, preferences later
I rely on AI-assisted coding	Medium	Simple architecture, a test for every tool, ship the MVP early

5 · PLAN AND SUCCESS CRITERIA

Milestones, and how I'll know it works

MILESTONES

Wk 1–2	Data tools: weather, astronomy, sites — each with tests
Wk 3	Single-user agent end to end → **midpoint demo**
Wk 4–5	Group coordination + day-of re-planning
Wk 6	UI polish, evaluation, README → **final demo**

Final demo: plan a real group trip for the Geminids (mid-December)

SUCCESS CRITERIA

- **Correct astronomy:** Moon and darkness times match published tables within a few minutes
- **Better nights:** on past weather, picked nights are clearer than random nights
- **Valid group plans:** 10 scripted groups, every hard constraint met
- **Real users:** stargazing friends try it and give feedback

APPENDIX · FEASIBILITY CHECK

Every data source already answers for Boston (tested Oct 25, 2026)

Source	What I got	Accuracy → how StarCrew handles it
Open-Meteo forecast	Hourly cloud cover by layer, 3 models side by side	Least certain input; models disagreed (Oct 26, 22:00: 8% vs 27%) → confidence level
7Timer! ASTRO	3-hourly seeing and transparency	Coarse grid → secondary signal only
Skyfield + JPL DE421	Moonrise/set, near-full Moon, full darkness from 19:19	Deterministic and precise → checked against published tables
CelesTrak ISS orbit	Two days of passes, each flagged sunlit or in shadow	Good for 1–2 days → refresh orbit data daily
Open-Meteo archive	Past hourly cloud cover (e.g. Dec 2025)	Ground truth for backtesting picks